www.tntextbooks.in

where $\frac{V_m}{R} = I_m$, the peak value of alternating current in the circuit. From equations (4.43) and (4.45), it is clear that the applied voltage and the current are in phase with each other in a resistive circuit. It means that they reach their maxima and minima simultaneously. This is indicated in the phasor diagram (Figure 4.46). The wave diagram also depicts that current is in phase with the applied voltage (Figure 4.46).

## 4.7.4 AC circuit containing only an inductor

Consider a circuit containing a pure inductor of inductance $L$ connected across an alternating voltage source (Figure 4.47). The alternating voltage is given by the equation.

$$\upsilon = V_m \sin \omega t \qquad (4.46)$$

The alternating current flowing through the inductor induces a self-induced emf or back emf in the circuit. The back emf is given by

$$\text{Back emf}, \varepsilon = -L\frac{di}{dt}$$

By applying Kirchoff's loop rule to the purely inductive circuit, we get

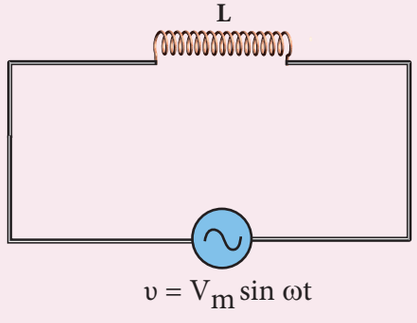



**Figure 4.47** AC circuit with inductance

$$\upsilon + \varepsilon = 0$$
$$V_m \sin \omega t = L\frac{di}{dt}$$
$$di = \frac{V_m}{L}\sin \omega t \; dt$$

Integrating both sides, we get

$$i = \frac{V_m}{L}\int \sin \omega t \, dt$$
$$i = \frac{V_m}{L\omega}\left(-\cos \omega t\right) + \text{constant}$$

The integration constant in the above equation is independent of time. Since the voltage in the circuit has only time dependent part, we can set the time independent part in the current (integration constant) into zero.

$$\left.\begin{aligned} &\cos \omega t = \sin\left(\pi/2 - \omega t\right) \\ &-\sin\left(\pi/2 - \omega t\right) = \sin\left(\omega t - \pi/2\right) \end{aligned}\right\}$$

$$i = \frac{V_m}{\omega L}\sin\left(\omega t - \pi/2\right)$$
$$\text{or} \quad i = I_m \sin\left(\omega t - \pi/2\right) \qquad (4.47)$$

where $\frac{V_m}{\omega L} = I_m$, the peak value of the alternating current in the circuit. From equation (4.46) and (4.47), it is evident that current lags behind the applied voltage by $\pi/2$ in an inductive circuit. This fact is depicted in the phasor diagram. In the wave diagram also, it is seen that current lags the voltage by 90° (Figure 4.48).

### Inductive reactance $X_L$

The peak value of current $I_m$ is given by $I_m = \frac{V_m}{\omega L}$. Let us compare this equation with $I_m = \frac{V_m}{R}$ from resistive circuit. The quantity $\omega L$ plays the same role as the resistance in resistive circuit. This is the resistance offered by the inductor, called inductive reactance ($X_L$). It is measured in ohm.

$$X_L = \omega L$$



UNIT-4(XII-Physics_Vol-1).indd 255 04-03-2019 10:40:52